Diagonalizing a matrix

A matrix can mix coordinates even when the underlying transformation acts by simple stretching. An eigenbasis reveals those directions. In that basis the transformation is diagonal, which turns repeated matrix multiplication into powers of scalars.

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1. Find the stretching directions

Definition
An eigenvector of A is a nonzero vector v satisfying $Av = \lambda v$. The scalar λ is its eigenvalue. An eigenbasis is a basis consisting of eigenvectors.

Being able to find one eigenvector tells us about a single direction. To describe every input using such directions, we need enough independent eigenvectors to span the space.

2. Change coordinates

Theorem
An $n \times n$ matrix over a field is diagonalizable over that field precisely when it has n linearly independent eigenvectors there.

Method
Put the eigenvectors in the columns of P. Put their eigenvalues in the same order on the diagonal of D. Column by column, $AP = PD$. Since the columns form a basis, P is invertible and $A = PDP^{-1}$.

Example
For $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, take $v_1 = (1, 0)^T$ and $v_2 = (1, 1)^T$. Multiplying gives $Av_1 = 2v_1$ and $Av_2 = 3v_2$. In this basis, $D = \operatorname{diag}(2, 3)$.

For a positive integer k, successive factors of $P^{-1}P$ cancel, leaving $A^k = PD^kP^{-1}$. Only the diagonal entries of D need to be raised to the power k.